CHAPTER 7

# Topos and Sheaf Theory

*"L'état de vérité partiel serait alors l'état de silence intérieur et d'écoute."*
– Alexander Grothendieck in La Clef des Songes

*"I do not approve. I understand."* – Spock to Anan 7

7.1. In the previous chapter, our intention was to introduce a few objects and mathematical notions like that of gluing (for example that of projective space from affine charts), local rings, and coordinate rings. In this chapter, our goal is to introduce the notion of topological spaces, shaves, ringed spaces and topoi which make up interesting part of EGA and SGA.

For reasons, we will bi-fold this chapter into Part I which will be about the local-to-global language of algebraic geometry and Part II which will the categorical viewpoint towards the notion space.

## Part I: The local-to-global language

7.2. The best point to start this local-to-global language is the *sheaf theory* which was precisely developed by Leray[1] [**32**] for analyzing the obstruction of piecing together local data to get global data.

### 1. Presheaf and Sheaf

Definition 7.3. Let $X$ be a topological space. We define presheaf $\mathcal{F}$ on $X$ with the assignment of $\mathcal{F}(U)$ to each open set $U \subset X$ such that for every inclusion of opens $V \subset U$, we have restriction map

$$\text{res}_U^V : \mathcal{F}(U) \to \mathcal{F}(V) \tag{195}$$

such that

This is a part of Prelude of Schemes project available at https://aayushayh.github.io/PtoS.html.
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[1]It is also interesting to note that Leray developed this 1946 while he was a prisoner prisoner of war in Oflag XVII-A in Austria during World War II.

(1) $\text{res}^U_U = id_{\mathcal{F}(U)}$ for all the opens $U$
(2) Whenever $W \subset V \subset U$, we have

$$\text{res}^W_V \cdot \text{res}^V_U = \text{res}^W_U \tag{196}$$

which is equivalent to saying that the following commute

$$\begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}(W) \\ & \searrow \quad \nearrow & \\ & \mathcal{F}(V) & \end{array}$$

Definition 7.4 (Morphism of presheaves). A morphism of presheaves $\mathcal{F}, \mathcal{G}$ on $X$ is a collection of maps $\alpha : \mathcal{F}(U) \to \mathcal{G}(U)$ for each open open set on $X$ such that whenever there is inclusion of opens sets, the following commutes

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\alpha(U)} & \mathcal{G}(U) \\ {\scriptstyle res^V_U}\downarrow & & \downarrow{\scriptstyle res^V_U} \\ \mathcal{F}(V) & \xrightarrow[\alpha(V)]{} & \mathcal{G}(V) \end{array}$$

whenever $V \subset U$.

7.5. In many places, one defines an element $f$ in $\mathcal{F}(U)$ to be the *section* over $U$. We can further simply $\text{res}^V_U(f)$ as $\sigma|_V$.

Definition 7.6 (Presheaf). In the language of categories, we can define a presheaf (of sets) on $\mathcal{C}$ to be a 'contravariant' functor from the category $\mathcal{C}$ (in our previous examples, $\mathcal{C}$ is the category of topological spaces) to the category of sets (again, it can be any other category).

The category of presheaves (of sets) is denoted by $PSh(\mathcal{C})_{Sets}$. The morphisms in this category are the natural transformations mentioned in Def. 7.4.

Definition 7.7 (Germs). Let $U, V \subset X$ and $\mathcal{F}$ be a presheaf. If we have two sections $f \in \mathcal{F}(U)$ and $g \in \mathcal{F}(V)$, and if they agree to an open neighbourhood $W \subseteq U \cap V$ through

$$f|_W = g|_W \tag{197}$$

then they are called germs.

In other words, two pairs $(f, U)$ and $(g, V)$ are equivalent are called germ at $p \in X$ if they satisfy (197) for an open neighborhood $W \subseteq U \cap V$ of a point $p$.

Definition 7.8 (Stalks). A stalk $\mathcal{F}_p$ at $p \in X$ is the collection of germs at $p$.

7.9. A very interesting method of studying a presheaf $\mathcal{F}$ is by studying the sections over smaller and smaller neighborhoods of a point $x \in X$.

DEFINITION 7.10 (One more definition of Stalks). We define the stalk at point $x$ as $\mathcal{F}_x$ to be $\xrightarrow{\text{limit}} \mathcal{F}(U)$ for $x \in U$ where $U's$ are the open neighborhood of $x$ which are partially ordered y the inclusion[2]

[2]The philosophy is that $\mathcal{F}_x$ is the disjoint union $\bigcup \mathcal{F}(U)$ of all the sections of $\mathcal{F}$ modulo equivalent sections (germs).

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